6D Object Pose Estimation via Deep Rotation Learning and Geometric Translation Recovery

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https://github.com/mauriziovergara/6D-pose-estimation

Abstract

This work addresses 6D object pose estimation from RGB images on the LineMOD benchmark. We design a two-stage pipeline that combines a YOLOv8-based detector for object localization with a dual-branch pose estimation network. The detector is fine-tuned on a pre-processed LineMOD split and achieves 0.995 mAP@0.5 and 0.935 mAP@[0.5:0.95] on the validation set, providing accurate bounding boxes for downstream pose prediction. The pose network employs a ResNet-50 backbone to regress a 4D rotation vector, subsequently normalized to a unit quaternion, and a lightweight CNN to regress depth (Z), while the remaining translation components (X, Y) are computed geometrically using the pinhole camera model with known camera intrinsics. Training uses a geodesic quaternion loss for rotation and an uncertainty-based multi-task loss weighting scheme to balance rotation and translation objectives. On the pose estimation task, our RGB-only model reaches a global mean ADD of 4.61 cm, with 21.04% of predictions within 2 cm, and a mean rotation error of 11.22° over 3,160 validation images across 13 LineMOD classes. Results demonstrate that a combination of a simple, geometrically constrained architecture with modern detection achieves competitive RGB-only performance, bypassing the need for depth maps or iterative refinement.

1. Introduction

Estimating the full 6D pose of rigid objects from images is a fundamental problem for robotic manipulation, augmented reality, and autonomous systems [10, 11]. While RGB-D methods such as DenseFusion [10] have demonstrated strong accuracy by fusing depth and color, many real-world settings still require robust RGB-only solutions due to hardware constraints, cost, or deployment simplicity. In this work we focus on 6D pose estimation from RGB on the LineMOD dataset. Our pipeline follows a two-stage design. First, a YOLOv8n detector localizes objects and produces axis-aligned bounding boxes in the image plane. Second, a ResNet-50–based PoseNet regresses a 4D rotation vector normalized to a unit quaternion and recovers translation by learning depth (Z) with a lightweight CNN while computing X,Y via the pinhole camera model from the bounding-box center and camera intrinsics. This design reduces the learning burden and enforces consistency with the camera geometry. Training uses a geodesic loss on $SO(3)$ for orientation, an L_1 loss for translation, and an automatic uncertainty-based weighting module (AWL) to balance the two tasks.

Our contributions are as follows:

- We present an end-to-end RGB-only 6D pose estimation pipeline on LineMOD that combines YOLOv8n detection with a ResNet-50-based PoseNet and evaluates under realistic detection noise from predicted bounding boxes.
- We introduce a geometric translation head that predicts object depth and reconstructs the full translation via the pinhole camera model, improving stability and interpretability of the pose predictions.
- We report a detailed per-class evaluation on LineMOD. Our model achieves a global mean ADD of 4.61 cm, a mean translation error of 4.46 cm, a mean rotation error of 11.22°, and 21.04% of poses within 2 cm, using only RGB inputs and YOLO-predicted bounding boxes.

2. Related Work

6D pose estimation from RGB. Early RGB-only methods often detect 2D keypoints and recover pose via PnP [3, 6]. More recent approaches rely on deep regressors that directly map image features to rotation and translation, sometimes focusing primarily on orientation estimation [11]. Large surveys highlight both the progress and remaining challenges of RGB-only 6D pose estimation, including robustness to occlusion and texture-less objects [5, 12].

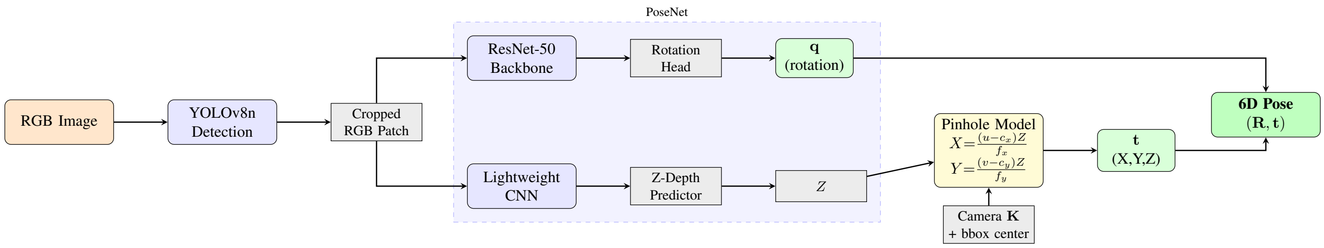

Figure 1. Overview of the proposed pipeline. An RGB image is processed by YOLOv8n to obtain object crops. PoseNet predicts rotation (quaternion) via ResNet-50 and Z-depth via a lightweight CNN. Translation (X, Y) is computed geometrically using the pinhole camera model.

RGB-D fusion. The availability of commodity depth sensors has motivated RGB-D approaches that explicitly exploit 3D geometry. PoseCNN [11] combines RGB segmentation with depth-based ICP refinement, while DenseFusion [10] jointly embeds RGB features and 3D point clouds at a per-pixel level and applies an iterative refinement network to improve pose estimates. These methods achieve high accuracy on benchmarks such as LineMOD and YCB-Video but require reliable depth maps and incur additional computational cost.

Geometric heads and camera models. Several works incorporate camera geometry into pose prediction to reduce the regression space. For instance, some architectures predict 2D projections or depth and reconstruct 3D locations using known intrinsics [3, 4]. Our translation head follows this philosophy: it predicts only depth and reconstructs the full translation (X, Y, Z) from the bounding-box center and the pinhole model, which encourages geometrically consistent poses while keeping the network purely RGB-based.

3. Method

3.1. Problem Formulation

Given an RGB image $I \in \mathbb{R}^{H \times W \times 3}$ and a set of known rigid object models, the goal is to estimate the 6D pose of each visible object instance. A pose is represented as a rigid transformation

$$T = (R, t), \quad R \in SO(3),\ t \in \mathbb{R}^3, \tag{1}$$

that maps points from the object coordinate system to the camera coordinate system, following the LineMOD convention. We parameterize R as a unit quaternion $q \in \mathbb{R}^4$ and t as a 3D translation vector. The overall pipeline, shown in Figure 1, consists of an object detector that predicts bounding boxes and a pose network that estimates (q, t) from cropped RGB patches.

3.2. Object Detection with YOLOv8n

We adopt the YOLOv8n model from the Ultralytics library as our object detector. The network is initialized

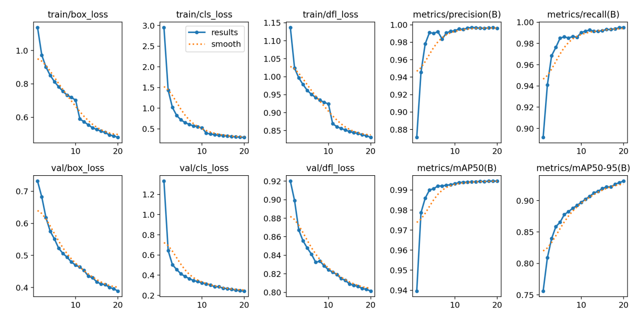

Figure 2. YOLOv8n training curves on LineMOD. Left: box loss converges rapidly. Right: mAP@0.5 reaches 0.995, mAP@[0.5:0.95] stabilizes at 0.935 by epoch 16.

from ImageNet-pretrained weights and fine-tuned on a preprocessed LineMOD dataset containing 15,800 RGB images across 13 classes. The data are split into 12,640 training and 3,160 validation images (80/20 split) and this partition is reused for pose estimation experiments. Training is configured for up to 30 epochs with a batch size of 32, although in practice optimization is stopped around epoch 16 when validation performance saturates.

On the 3,160-image validation set, YOLOv8n achieves 0.995 mAP@0.5 and 0.935 mAP@[0.5:0.95], indicating highly accurate bounding boxes suitable for precise cropping. Figure 2 shows the training curves, while Figure 3 displays a representative training batch demonstrating robust object detection across diverse objects and occlusion patterns. During pose training and evaluation, we always use YOLO-predicted bounding boxes rather than ground-truth boxes, making the pipeline robust to realistic detection noise. The predictions are serialized into a YAML file and later consumed by the pose dataset loader.

3.3. Pose Dataset and Preprocessing

For pose estimation we implement a custom `LineModDataset` that parses the LineMOD directory structure. For each class, the loader reads the ground-truth annotations from `gt.yml`, which contain the rotation

Figure 3. Representative YOLOv8n detections during training on cluttered LineMOD scenes. High-confidence bounding boxes ($\geq$25%) enable precise cropping despite occlusions and varying appearances.

matrix $\mathrm{cam\_R\_m2c}$, the translation vector $\mathrm{cam\_t\_m2c}$, and the 2D bounding box $\mathrm{obj\_bb}$. Translations are converted from millimetres to metres, and rotations are converted from matrices to quaternions using SciPy, re-ordered to the (w, x, y, z) convention and normalized.

Given an image and its corresponding YOLO prediction, we crop the RGB patch within the predicted bounding box. If the prediction is missing or invalid, we fall back to the ground-truth box to avoid discarding samples. YOLO inference employs a 25% confidence threshold, which may lead to occasional missed detections. Given the medium-sized LineMOD dataset, this fallback mechanism preserves samples without compromising PoseNet performance, as it affects at most a few instances. The crop is padded to a square aspect ratio, resized to 224×224, and transformed with simple photometric augmentations during training (colour jitter and normalization) and with deterministic resizing during validation. For each sample the dataset returns the cropped RGB tensor, the ground-truth translation t, the ground-truth quaternion q, the 2D coordinates of the bounding-box centre (u, v), the camera intrinsic matrix K, and the class id.

3.4. PoseNet with Geometric Translation Head

Network architecture. Our pose network, `PoseNetRGBGeometric`, consists of two branches for rotation and translation. The rotation branch uses a ResNet-50 backbone pre-trained on ImageNet. We remove the final classification layer and apply global average pooling, followed by a multi-layer perceptron with dimensions $2048 \rightarrow 1024 \rightarrow 512 \rightarrow 4$, BatchNorm, ReLU, and Dropout. The output quaternion $\hat{q}$ is normalized to unit length to enforce $\hat{q} \in S^3$. During training, we freeze 70% of the ResNet-50 backbone to preserve pre-trained features while allowing the rotation head to adapt to the pose estimation task. This selective freezing helps prevent catastrophic forgetting of ImageNet features while allowing task-specific adaptation in the decoder layers.

The translation branch is designed to predict depth and reconstruct the full translation from camera geometry. A lightweight convolutional backbone processes the same 224×224 RGB crop and applies four convolutional blocks with BatchNorm, ReLU, max pooling, and final adaptive average pooling, resulting in a 256-dimensional feature vector. This vector is passed to a small MLP with layers $256 \rightarrow 128 \rightarrow 64 \rightarrow 1$ that outputs a scalar depth $\hat{Z}$. The last linear layer bias is initialized to 0.5 m to match typical object distances in LineMOD.

Geometric translation via pinhole camera. Given the predicted depth $\hat{Z}$, the bounding-box centre (u, v), and the intrinsic matrix

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{2}$$

we reconstruct the translation components X and Y via the pinhole camera model:

$$\hat{X} = (u - c_x)\frac{\hat{Z}}{f_x}, \quad \hat{Y} = (v - c_y)\frac{\hat{Z}}{f_y}. \tag{3}$$

The predicted translation is then

$$\hat{t} = (\hat{X}, \hat{Y}, \hat{Z})^\top. \tag{4}$$

If K or (u, v) are unavailable, the network falls back to a mode where only $\hat{Z}$ is predicted and $(\hat{X}, \hat{Y})$ are set to zero, but this configuration is not used in our main experiments.

3.5. Training Objective

Training is formulated as a multi-task regression problem over rotation and translation. Our training framework implements both Chordal and Geodesic metrics for the rotational component. Initial experiments were conducted using the Chordal loss; however, to address persistent angular errors and refine the estimation precision, we transitioned to a Geodesic loss on the $SO(3)$ manifold, which yielded superior convergence. Let $\hat{q}$ and q denote the predicted and ground-truth quaternions, both normalized to unit length. The geodesic rotation loss is defined as

$$\mathcal{L}_{\mathrm{rot}} = 2 \arccos\left(|\langle \hat{q}, q \rangle|\right), \tag{5}$$

where the inner product is clamped to avoid numerical instabilities near ± 1. This loss directly minimizes the angular error in radians.

For translation we use an L_1 loss:

$$\mathcal{L}_{\text{trans}} = \left\| \hat{t} - t \right\|_1 . \tag{6}$$

To automatically balance the two tasks we employ an uncertainty-based weighting scheme following Kendall loss. Let σ_{rot} and σ_{trans} denote learnable task uncertainties, implemented as log-variance parameters. The final loss is

$$\mathcal{L} = \frac{\lambda_{\text{rot}} \mathcal{L}_{\text{rot}}}{2\sigma_{\text{rot}}^2} + \frac{\lambda_{\text{trans}} \mathcal{L}_{\text{trans}}}{2\sigma_{\text{trans}}^2} + \log \sigma_{\text{rot}} + \log \sigma_{\text{trans}}. \tag{7}$$

where λ_{rot} and λ_{trans} are scalar weights that default to 1. This formulation allows the model to adaptively adjust the effective contribution of each task during training. The introduction of λ_{rot} and λ_{trans} provides a mechanism to manually balance the relative importance of rotation and translation losses during training. In practice, when the rotation loss formulation was changed from a chordal distance to a geodesic distance, λ_{rot} was temporarily increased to accelerate the convergence of the rotation component, while λ_{trans} was kept fixed. Once stable convergence was achieved, both λ_{rot} and λ_{trans} were reset to 1, allowing the automatic weighting scheme to operate without manual intervention. This design choice enhances the robustness and flexibility of the loss function in multi-task learning scenarios.

3.6. Optimization Details

We train PoseNet using the AdamW optimizer on the union of network parameters and uncertainty parameters, with a learning rate of 1×10^{-4}, weight decay of 1×10^{-6}, and batch size 32. These hyperparameters can be adapted during training to accommodate task-specific requirements and prevent overfitting. A configurable freeze ratio can be applied to the ResNet-50 backbone to keep early layers fixed in initial epochs and later unfreeze them; in our experiments, we maintain a constant freeze ratio of 70% throughout training, preserving the majority of pre-trained ImageNet features while allowing the final decoder layers to adapt to the pose estimation task. ReduceLROnPlateau scheduler monitors the validation loss and halves the learning rate when no improvement is observed for a specified patience period. The best checkpoint is selected according to the lowest mean ADD on the validation set, and early stopping can terminate training if ADD does not improve for several consecutive epochs.

4. Experiments

4.1. Ablation Study

To validate the proposed dual-branch architecture, we first trained a baseline PoseNet with a single branch directly regressing quaternion rotation and full translation (x, y, z) from a frozen ResNet-50 backbone, as shown in Figure 4. This direct regression approach yielded mean ADD=10.29 cm with only 0.9% success rate at the stringent ADD<2 cm threshold. This performance was insufficient for our strict accuracy requirements, motivating the dual-branch design that achieves 4.61 cm ADD and 21.04% ADD<2 cm. These results are reported in Table 1.

Figure 4. Baseline architecture (single head direct regression). Frozen ResNet-50 regresses quaternion and full (x, y, z) translation, yielding poor results (ADD=10.29 cm).

Method	ADD (cm)	ADD<2cm
Baseline (single head)	10.29	0.9%
Ours (dual head)	**4.61**	**21.04%**

Table 1. Baseline vs proposed method on LineMOD validation.

The poor performance highlights the difficulty of learning geometric constraints implicitly. Our dual-head design—rotation via quaternion regression and translation via explicit pinhole model—provides strong inductive bias, reducing mean ADD by ∼55% (from 10.29 cm to 4.61 cm).

4.2. Dataset and Metrics

We evaluate the proposed method on the LineMOD dataset, using a pre-processed version containing 15,800 RGB images across 13 object categories. The data are split into 12,640 training and 3,160 validation images following an 80/20 ratio with a seed = 42 for reproducibility. This split is used consistently for YOLO training and inference, and pose training and validation.

To measure pose accuracy we follow common practice and use the Average Distance of Model Points (ADD) metric [2, 10]. Let $\{x_j\}_{j=1}^{M}$ denote a set of 3D model points, and let (R, t) and $(\hat{R}, \hat{t})$ denote the ground-truth and predicted poses, respectively. The ADD for one sample is

$$\text{ADD} = \frac{1}{M} \sum_{j=1}^{M} \left\| \hat{R} x_j + \hat{t} - (R x_j + t) \right\|_2 . \tag{8}$$

Model points are obtained by loading the official LineMOD meshes and normalizing units to metres.

4.3. Training Configuration

YOLOv8n is fine-tuned for up to 30 epochs with batch size 32 on the training split. In practice, validation mAP saturates early and training is stopped manually around epoch 16. On the 3,160-image validation set the detector achieves approximately 0.995 mAP@0.5 and 0.935 mAP@[0.5:0.95], averaged over all classes. The resulting bounding boxes are saved to a YAML file and later used by the pose dataset loader.

PoseNet is trained on crops generated from YOLO-predicted bounding boxes. We employ the geodesic rotation loss from Eq. (5), the translation loss from Eq. (6), and uncertainty-based weighting from Eq. (7). The optimizer is AdamW with an initial learning rate of 1×10^{-4}, weight decay of 1×10^{-6}, and a ReduceLROnPlateau scheduler that decreases the learning rate upon validation plateaus; the learning rate was further manually reduced to 5×10^{-6} when needed. The weight decay was increased to 3×10^{-5} to mitigate overfitting. The batch size is fixed at 32 and two data-loader workers are used. Training proceeds for up to 110 epochs, and the best checkpoint, selected by mean validation ADD, is reached at epoch 79.

Hyperparameter	YOLOv8n	PoseNet
Learning rate	0.01	$1 \times 10^{-4} \rightarrow 5 \times 10^{-6}$
Weight decay	5×10^{-4}	$1 \times 10^{-6} \rightarrow 3 \times 10^{-5}$
Batch size	32	32
Epochs	30 (stop @16)	110 (best @79)
Optimizer	SGD	AdamW
Scheduler	Linear LR	ReduceLROnPlateau
Rotation loss	-	Chordal or Geodesic
Translation loss	-	L1
Task weighting	Default	Automatic Weighting Loss
Backbone freeze	-	ResNet-50 (70%)
Data augment.	Default	ColorJitter
Image size	640×640	224×224
Confidence thresh.	25%	-
Val. mAP@0.5	0.995	-
Val. mean ADD	-	4.61 cm

Table 2. Hyperparameters and training configuration.

4.4. Quantitative Results

Figure 5 reports the per-class mean ADD (cm) on the preprocessed LineMOD validation set, highlighting non-uniform performance across object categories. This variability is consistent with class-dependent factors such as object geometry (e.g. surface distinctiveness) and the degree of occlusion present in the evaluation images. Table 3 reports the corresponding per-class results of PoseNet, including the $\mathrm{ADD} < 2\,\mathrm{cm}$ and $\mathrm{ADD-S} < 2\,\mathrm{cm}$ success rate (relevant for grasping), mean ADD and ADD-S. The last row summarizes the global averages over all 3,160 validation images.

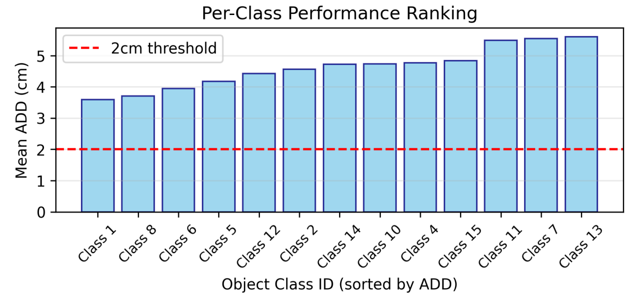

Figure 5. Per-class ADD ranking. Class 1 (3.59cm) performs best; class 13 (5.60cm) is most challenging due to occlusion/texture.

Class	%ADD<2cm	$ADD_m[cm]$	%ADD-S	$ADD\text{-}S_m[cm]$
1	37.99	3.59	68.12	1.86
2	16.06	4.56	63.05	1.90
4	13.31	4.77	53.99	2.21
5	27.35	4.17	64.96	1.89
6	30.12	3.94	68.27	1.81
8	10.92	5.54	48.91	2.79
9	28.14	3.70	69.96	1.85
10	25.21	4.73	64.29	2.26
11	21.90	5.49	51.82	3.13
12	20.25	4.42	65.70	2.02
13	6.60	5.60	49.53	2.65
14	17.50	4.72	59.17	2.16
15	16.81	4.83	57.98	2.37
Mean	**21.04**	**4.61**	**60.51**	**2.22**

Table 3. Complete per-class results on LineMOD validation set (ADD/ADD-S in cm).

Globally the model attains an average ADD of 4.61 cm with 21.04% of predictions below the 2 cm threshold, a mean translation error of 4.46 cm, and a mean rotation error of 11.22°, as shown in Table 4. Performance varies across objects: class 1 reaches 37.99% within 2 cm and 3.59 cm mean ADD, while class 13 reaches only 6.60% within 2 cm and 5.60 cm mean ADD, reflecting differences in appearance, geometry, and occlusion patterns.

These results are obtained without using depth information, explicit symmetry handling, or iterative refinement such as ICP [4, 10, 11]. The pose network operates solely on RGB crops derived from YOLO detections and must therefore tolerate small misalignments in the bounding boxes. Despite this, the global errors remain moderate, indicating that both the geometric translation and rotation heads provide a strong inductive bias for 6D pose regression. Figure 6 demonstrates a qualitative result from our pipeline applied to LineMOD object 04 (camera). YOLOv8n provides the object detection (cyan bounding box), while PoseNetRGB-Geometric predicts the 6D pose. The close alignment between predicted axes and object orientation indicates a low pose error of a few degrees, consistent with quantitative results (ADD=4.77 cm for this class).

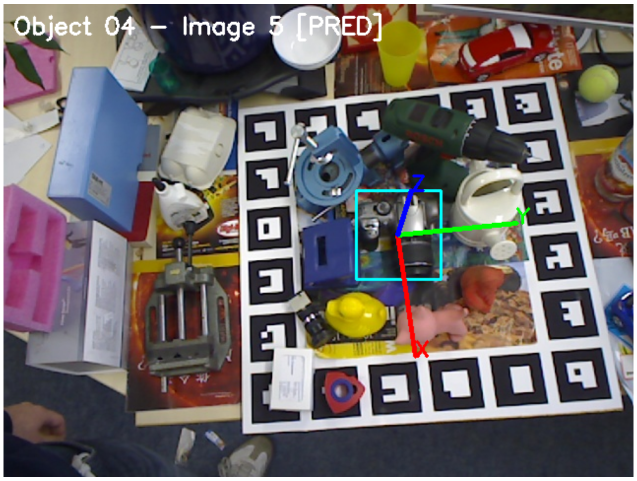

Figure 6. Example qualitative result from our pipeline on LineMOD object 04. The cyan bounding box is predicted by YOLOv8n; red/green/blue axes show the predicted 6D pose from PoseNet (rotation via quaternion head, translation via geometric pinhole model). The alignment indicates a low error of a few degrees.

Table 5 compares our method against DenseFusion [10] (RGB-D) and two RGB-only baselines SSD-6D [3] and Real-Time Seamless Single Shot 6D Object Pose Prediction [9] using the standard $0.1D$ threshold with ADD for the non-symmetric objects and ADD-S for symmetric objects (Eggbox: class 10; Glue: class 11), making results directly comparable. DenseFusion fuses RGB and depth with iterative refinement (per-pixel 86.2% $\rightarrow$ iterative 94.3%), leveraging multi-modal inputs and optimization unavailable to RGB-only methods. Our pipeline achieves 31.7% global accuracy—competitive with RGB-only SOTA (55.95% RSS-SSP)—using a single forward pass, YOLOv8n-detected boxes, and no refinement, making it dramatically lighter for real-time deployment while approaching RGB-D performance without depth sensors.

Notably, our mean rotation error ($\mathrm{Rot}_m = 11.22°$, 9 classes $< 12°$) satisfies robotics requirements, where orientation precision often dominates grasping success.

5. Conclusion

We presented an RGB-only 6D object pose estimation pipeline for the LineMOD dataset combining YOLOv8n detection with a dual-branch PoseNet built on a ResNet-50 backbone featuring a rotation head and a lightweight CNN with a geometric translation head. The detector delivers accurate bounding boxes (0.995 mAP@0.5), while PoseNet regresses orientation as a unit quaternion and predicts Z-depth from cropped RGB patches; the full 3D translation is then reconstructed from the predicted Z-depth and the bounding box center using the pinhole camera model. Employing a chordal rotation loss followed by a geodesic rotation loss, together with uncertainty-based multi-task weighting, our approach achieves a mean ADD of 4.61 cm, 31.7% ADD@0.1D accuracy, translation error of 4.46 cm, rotation error of 11.22°, and 21.04% of poses within 2 cm across the 3,160-image validation set, relying solely on RGB inputs and YOLO-predicted boxes. These results demonstrate that simple, geometrically constrained architectures yield competitive RGB-only performance without depth sensing or iterative refinement, establishing a strong baseline for future RGB-D fusion or refinement extensions

Class	$Trans_m$ (cm)	Rot_m (°)
1	3.47	14.08
2	4.44	10.24
4	4.64	10.24
5	3.94	10.96
6	3.84	10.03
8	5.29	11.33
9	3.61	12.77
10	4.51	13.09
11	5.41	10.37
12	4.29	12.42
13	5.41	11.02
14	4.48	9.74
15	4.71	9.77
Mean	**4.46**	**11.22**

Table 4. Per-class translation and rotation errors on LineMOD validation set.

Object	DenseFusion (RGB-D)		RGB-only w/o refinement		
	per-pixel	iterative	SSD-6D	RSS-SSP	Ours
ape	79.5	92.3	0	21.62	21.8
bench vi.	84.2	93.2	0.18	81.80	38.2
camera	76.5	94.4	0.45	68.57	18.6
can	86.6	93.1	1.35	68.80	41.9
cat	88.8	96.5	2.58	41.52	28.9
driller	77.7	87.0	5.18	63.51	28.0
duck	76.3	92.3	0.58	27.23	16.7
eggbox	99.9	99.8	8.9	69.58	63.9
glue	99.4	100.0	0	80.02	51.1
hole p.	79.0	92.1	0.30	42.63	15.3
iron	92.1	97.0	8.86	74.97	20.8
lamp	92.3	95.3	8.20	71.11	37.5
phone	88.0	92.8	0.18	47.74	29.4
Mean	**86.2**	**94.3**	**2.42**	**55.95**	**31.7**

Table 5. Per-class comparison on LineMOD using the standard $0.1\ D$ ADD/ADD-S accuracy. DenseFusion (RGB-D with refinement) is compared against RGB-only methods without refinement. OURS achieves competitive results among RGB-only approaches.

such as DenseFusion [10].

5.1. Future work

Beyond the LineMOD benchmark, the proposed RGB-only architecture is directly applicable to automotive scenarios such as pose estimation of tools, components, or infrastructure elements from monocular cameras in assembly lines and driver-assistance systems. Extending the pipeline to multi-object scenes with heavy occlusions and object interactions, multi-view settings and to outdoor datasets would enable integration into perception stacks for autonomous driving and robotic manipulation around vehicles.

Recent works demonstrate 6D augmentation [7] and contour alignment refinement [1] boost LineMOD accuracy 5-10%. In future work, performance could be further improved by reducing input noise, for instance through background removal, instance-level cropping, and photometric normalization to better isolate the target object and regularize illumination. Recent noise-elimination approaches for 6D pose estimation, such as Uni6Dv2, adopt a two-stage denoising pipeline in which an instance segmentation network suppresses instance-outside background pixels, followed by a lightweight depth denoising module that calibrates depth features before pose regression, resulting in improved ADD(S) performance on LineMOD [8]. Such preprocessing is particularly relevant in robotic and automotive settings, where sensor artifacts, motion blur, and cluttered backgrounds can degrade pose accuracy.

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